

FIN Programs P497–P506

Localization, Certified Stability, Phase Topology,
and Identifiability

A bounded analytical and small-compute continuation of Release 10.49

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Scope. This report executes all ten low-compute recommendations P497–P506 from Release 10.49. The stopped P485/P487 Gröbner campaign and global exact complex-cone problems remain excluded.

Evidence boundary. Analytic theorems, outward interval evaluation, finite numerical continuation, FFT quadrature and synthetic records are admitted. A supplied nonlinear law is never promoted to a consequence of FIN. No laboratory or external-audit evidence is claimed.

Runtime. The complete deterministic batch ran locally in approximately 21 seconds. Twelve regression and certificate tests passed.

Repository. <https://github.com/hyconiek/Fractal-Nadsoliton-Theory>

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Abstract

Ten bounded FIN studies are executed. The principal result is conditional but mathematically substantive. For the supplied focusing discrete nonlinear Schrödinger law

$$i\dot{\psi} = \kappa A\psi - |\psi|^2\psi,$$

the strict $\mathbb{Z}_{12}$ generator admits a one-site anti-continuum state whose stationary Jacobian is exactly $\text{diag}(-2, 1, \dots, 1)$. The implicit-function theorem therefore produces a unique local real branch. Numerical continuation reaches the full strict coupling $\kappa = 1$ without a break and with residual 2.22×10^{-16} , inverse participation ratio 0.654578, and central-site power fraction 0.804496. This is a localized nonlinear state, unlike the delocalized cycles of P488. It is not yet a FIN-derived physical nadsoliton because the focusing cubic law and frequency are supplied.

The P488 spectral stability calculation is reduced analytically to 2×2 Fourier blocks and evaluated by outward interval arithmetic. Every nonzero real exponent is strictly negative in the declared decimal strict model; mode three retains an additional exact neutral block. A two-frequency clock theorem proves periodicity precisely for rational frequency ratio and injectivity for irrational ratio. For the strict first two frequencies, rationals with denominator at most 10^6 are excluded by an outward interval certificate, but global irrationality remains open.

The Stieltjes inverse problem receives a local rational-left-inverse certificate. It proves local injectivity only inside a log-parameter box of radius 10^{-9} and for response noise below 1.17×10^{-16} , quantitatively confirming severe practical fragility. The even/odd context family is extended through C_{192} and C_{384} and compared with a polyphase integral fingerprint; the C_{384} discrepancy is 0.003244 and decreases monotonically after C_{96} .

The Legacy*-strict linear parameter homotopy has two distinct boundaries. Its signed Laplacian becomes positive semidefinite near $t = 0.7008185569$, whereas all six radial weights become nonnegative only at the exact $d = 6$ phase crossing $t = 0.9257900390$. A nonvanishing S^1 phase field with branch-cut margin is then constructed; bounded updates and geodesic refinement preserve integer winding exactly. Finally, one context from the P495 synthetic catalogue suffices to distinguish its four models, while a trajectory no-go proves that no finite set of kernel observations can identify a unique continuous parameter evolution law.

The batch therefore supplies a credible conditional localization mechanism, completes several certification tasks, and sharpens the missing-law boundary. It does not derive the cubic nonlinearity, absolute units, a selector, the legacy-to-strict parameter source, laboratory physics, particles, or a Theory of Everything.

Status vocabulary. **[PROVEN]** means an analytic statement or exact finite proof in its declared assumptions; **[STRONG EVIDENCE]** means robust numerical or synthetic evidence; **[CONDITIONAL]** marks dependence on a supplied dynamical or operational law; **[MODERATE EVIDENCE]** marks bounded evidence that is not a global theorem; **[REFUTED]** marks a scoped no-go or counterexample.

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1 Executive results

Principal advance. P497 crosses the localization boundary left open by P488. A focusing strict-DNLS branch exists rigorously near zero coupling and remains strongly localized in numerical continuation to full strict coupling. The advance concerns the mathematical capability of the strict generator after a cubic law is supplied; it does not derive that law from FIN.

Table 1: Programs, typed objects, and terminal verdicts.

ID	New object	Main result	Status
P497	O201 strict-DNLS localization branch	Local IFT theorem and full-coupling numerical localization, IPR 0.654578.	[CONDITIONAL] / [STRONG EVIDENCE] [PROVEN]
P498	O202 Fourier-block stability enclosure	All nonzero P488 real exponents are interval-enclosed below zero; one extra neutral block for $m = 3$.	
P499	O203 two-frequency torus clock	Rational/irrational dichotomy proven; no rational with denominator $\leq 10^6$ for the strict interval.	[PROVEN] / [MODERATE EVIDENCE]
P500	O204 Stieltjes inverse box	Local inverse certified, but only at extremely small noise and parameter radius.	[CONDITIONAL]
P501	O205 polyphase integral fingerprint	C_{192}, C_{384} approach a declared asymptotic integral fingerprint.	[STRONG EVIDENCE]
P502	O206 passivity/PSD boundary ledger	Exact radial boundary 0.9257900390 differs from PSD boundary 0.7008185569.	[PROVEN] / [STRONG EVIDENCE]
P503	O207 nonvanishing phase-refinement sector	Bounded updates and midpoint refinement preserve winding.	[PROVEN]
P504	O208 one-context coherence tester	One catalogue context separates four synthetic models with 99.925% simulated accuracy.	[STRONG EVIDENCE]
P505	O209 neutral theorem interface	Six theorem nodes pass schema and dependency-DAG validation.	[PROVEN] in interface scope
P506	O210 trajectory observation quotient	Static parameters are locally observable; finite records never identify a unique continuous path.	[PROVEN] / [STRONG EVIDENCE]

2 Shared strict finite core and kernel discipline

On $V = \mathbb{Z}/12\mathbb{Z}$, let

$$W_{xy} = \frac{\cos(0.18575d(x, y) + 0.16250)}{1 + d(x, y)^{1.8}}, \quad A = \text{diag}(W\mathbf{1}) - W.$$

The row sum of W is $1.6603072787660986\dots$, all six strict radial weights are positive, and $A \succeq 0$. This report uses A only as the later strict working generator. It does not identify it silently with

$$K_{\text{legacy}}(d) = 4 \ln 2 \frac{\cos(\pi d/4 + \pi/6)}{1 + 0.01d}$$

or transfer any legacy electromagnetic, hierarchy, Planck-scale, or matter role.

3 P497 — localized nonlinear strict states

3.1 The supplied DNLS object

Consider the focusing finite DNLS equation

$$i\dot{\psi} = \kappa A\psi - g|\psi|^2\psi, \text{quad}g = 1, \text{quad}0 \leq \kappa \leq 1. \quad (1)$$

With $\psi(t) = e^{-i\mu t}u$, $\mu = -1$ and real u , the stationary equation is

$$F(u, \kappa) = \kappa Au - u^{\circ 3} - \mu u = 0. \quad (2)$$

Theorem 3.1 (Anti-continuum persistence). *At $\kappa = 0$, the state $u_0 = (1, 0, \dots, 0)$ solves (2). There is a unique real stationary branch through u_0 for all sufficiently small $|\kappa|$.*

Proof. At $(u_0, 0)$,

$$D_u F = -3 \operatorname{diag}(u_0^2) - \mu I = \operatorname{diag}(-2, 1, \dots, 1).$$

It is invertible, so the finite-dimensional implicit-function theorem applies. $\square$

This is the theorem-level part of P497. The extension from a nonzero neighbourhood of $\kappa = 0$ to $\kappa = 1$ is a numerical continuation result, not a global continuation proof.

Table 2: Selected points on the continued branch.

κ	power	IPR	peak fraction	$\ F\ _\infty$
0.00	1.000000	1.000000	1.000000	0
0.25	1.420032	0.954169	0.976737	2.22×10^{-16}
0.50	1.863852	0.863450	0.928550	2.22×10^{-16}
0.75	2.346621	0.759523	0.869404	2.22×10^{-16}
1.00	2.881677	0.654578	0.804496	2.22×10^{-16}

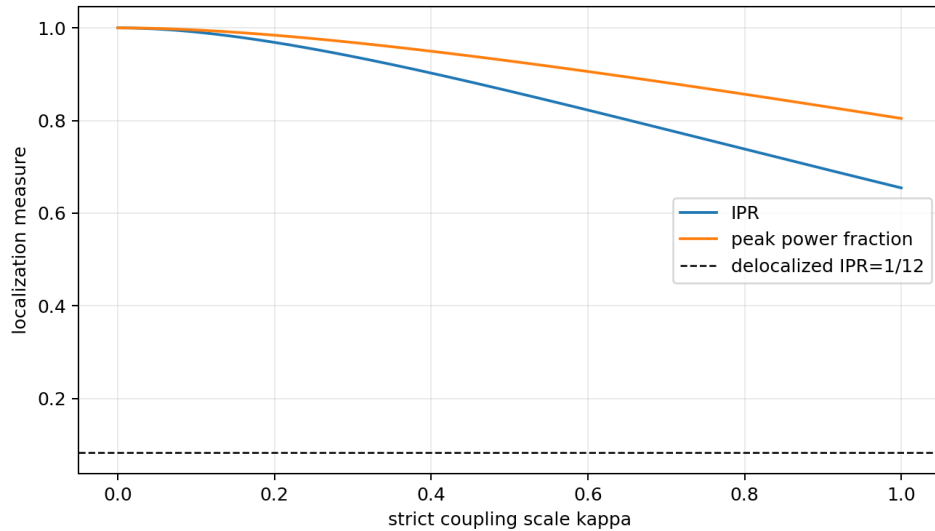


Figure 1: Localization decreases continuously as strict coupling is turned on, but remains far above the delocalized value $1/12$ at $\kappa = 1$.

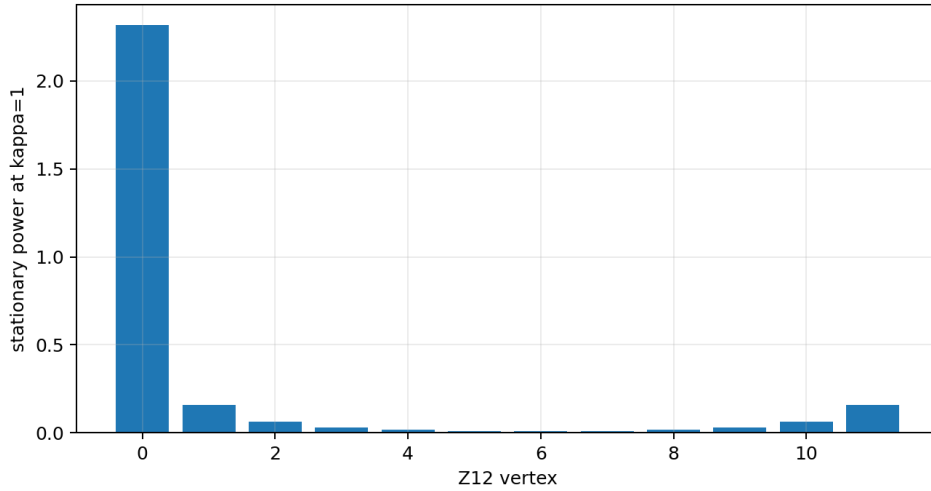


Figure 2: Power profile at full strict coupling. The state is pinned at the selected vertex; translation produces eleven equivalent copies.

The Hamiltonian linearization has no resolved real exponent above 10^{-6} along the sampled branch, but near-zero Hamiltonian eigenvalues require a separate orbital-stability certificate. Therefore asymptotic stability is not claimed. Nor is mobility: a pinned discrete breather is not yet a travelling soliton.

P497 verdict. [CONDITIONAL] local persistence is proven for the supplied DNLS; [STRONG EVIDENCE] full-coupling localization is numerically established. The source and sign of the cubic law, the choice $\mu = -1$, continuum persistence, orbital stability and physical interpretation remain open.

4 P498 — interval certification of P488 stability

Let λ_j be the strict Fourier eigenvalues and let m label a persistent P488 background. Fourier perturbations with offsets q and $-q$ reduce the real 24×24 linearization to 2×2 blocks. If

$$d_{m,q} = \frac{\lambda_{m+q} + \lambda_{m-q} - 2\lambda_m}{2},$$

then the two block exponents have real parts

$$\Re \rho_{m,q}^{\pm} = -r \pm \Re \sqrt{r^2 - d_{m,q}^2}, \quad r = 0.2. \quad (3)$$

All strict weights and eigenvalues were evaluated with 70-decimal outward interval arithmetic.

Table 3: Largest certified upper bound among nonzero real parts for each excited background.

mode m	largest nonzero upper bound	exact neutral offsets
1	−0.0029812392	$q = 0$
2	−0.0065292023	$q = 0$
3	−0.0138368211	$q = 0, 6$
4	−0.0124870968	$q = 0$
5	−0.0019318268	$q = 0$
6	−0.0048048352	$q = 0$

The $q = 0$ block is global phase neutrality. For $m = 3, q = 6$, inversion symmetry gives $\lambda_{m+q} = \lambda_{m-q} = \lambda_m$, so the second neutrality is exact rather than a rounding accident. P498 upgrades P488’s nonzero transverse signs, but it correctly refuses asymptotic stability for mode three because of that additional neutral direction.

5 P499 — two-frequency clock atlas

Theorem 5.1 (Torus clock dichotomy). *For nonzero ω_1, ω_2 , the map*

$$t \mapsto (e^{-i\omega_1 t}, e^{-i\omega_2 t}) \in \mathbb{T}^2$$

has a nonzero period if and only if $\omega_2/\omega_1 \in \mathbb{Q}$. If the ratio is irrational, the map is injective on $\mathbb{R}$.

Proof. A repeated value at times t and $t + T$ requires both $\omega_1 T$ and $\omega_2 T$ to be integer multiples of 2π . A nonzero common T exists exactly when the frequency ratio is rational. $\square$

For the strict modes,

$$\omega_1 = 0.7541211542070798, \quad \omega_2 = 1.5770495144276093,$$

and outward interval evaluation gives

$$2.0912415805200384 \leq \omega_2/\omega_1 \leq 2.0912415805200393.$$

For every $1 \leq q \leq 10^6$, the interval $q[\omega_2/\omega_1]$ excludes all integers. The smallest paid separation is 1.2852×10^{-7} at $q = 740693$. Thus no rational with denominator at most 10^6 is possible. This is not a proof of irrationality.

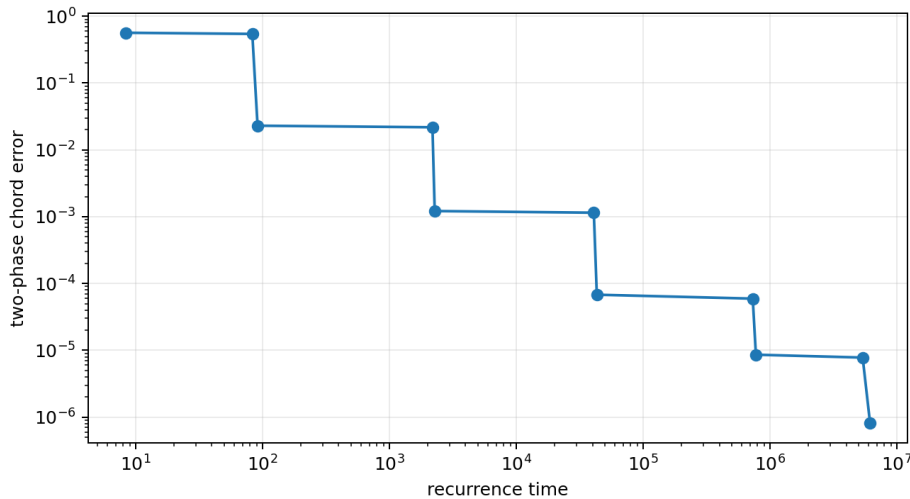


Figure 3: Continued-fraction recurrence quality. Small phase error is purchased with a rapidly increasing dimensionless waiting time.

The best audited recurrence has dimensionless time 6.1713×10^6 and two-phase chord error 8.09×10^{-7} . Like P489, this clock is dimensionless and invariant under common generator/time rescaling.

6 P500 — Stieltjes identifiability bounds

The three-pole response is

$$\sigma(z; \theta) = \sum_{j=1}^3 \frac{e^{b_j}}{z + e^{a_j}}, \quad \theta = (a_1, a_2, a_3, b_1, b_2, b_3).$$

Let F be the 48-component relative response map at the P490 sampling points. A rational approximation R to the left inverse of $J_0 = DF(\theta_*)$ was constructed with denominator at most 10^{12} . Interval evaluation on the box $\|\theta - \theta_*\|_\infty \leq 10^{-9}$ gives

$$\|RDF(\theta) - I\|_F \leq q = 0.245562333 < 1.$$

Proposition 6.1 (Local inverse bound). *Throughout this box,*

$$\|F(\theta) - F(\theta_*)\|_2 \geq c\|\theta - \theta_*\|_2, \text{quad}c = \frac{1-q}{\|R\|_F} = 1.17160 \times 10^{-7}.$$

Proof. Integrate DF on the segment from θ_* to θ , multiply by R , and use the triangle inequality with $\|RDF - I\|_2 \leq \|RDF - I\|_F \leq q$ and $\|Rv\|_2 \leq \|R\|_F\|v\|_2$. $\square$

The self-consistency condition requires response-noise norm below

$$c \times 10^{-9} = 1.17160 \times 10^{-16}.$$

The condition number is 3.4729×10^7 . Thus local uniqueness exists in a narrow mathematical sense, while ordinary finite-precision pole recovery is fragile. The certificate is conditional on the declared rounded P490 poles, residues, contexts and response definition.

7 P501 — context-family limit

For the summable attenuation envelope $h_d = (1 + d^{1.8})^{-1}$, split the infinite lattice into even and odd sites. Let $C(\theta)$ be the odd–odd Laplacian symbol plus $\delta = 0.1$, and $B(\theta)$ the even–odd symbol. The limiting trace response is

$$\sigma(z) = \frac{1}{2\pi} \int_0^{2\pi} \frac{|B(\theta)|^2}{z + C(\theta)} d\theta. \quad (4)$$

The formula follows by diagonalizing the two polyphase convolution blocks with the Fourier transform. It supplies a natural continuum-context object; it does not supply exact finite self-similarity.

Table 4: Maximum normalized-fingerprint discrepancy from the FFT approximation of (4).

cycle	maximum discrepancy
C_{12}	0.0372626
C_{24}	0.0312736
C_{48}	0.0183403
C_{96}	0.0102133
C_{192}	0.00574028
C_{384}	0.00324399

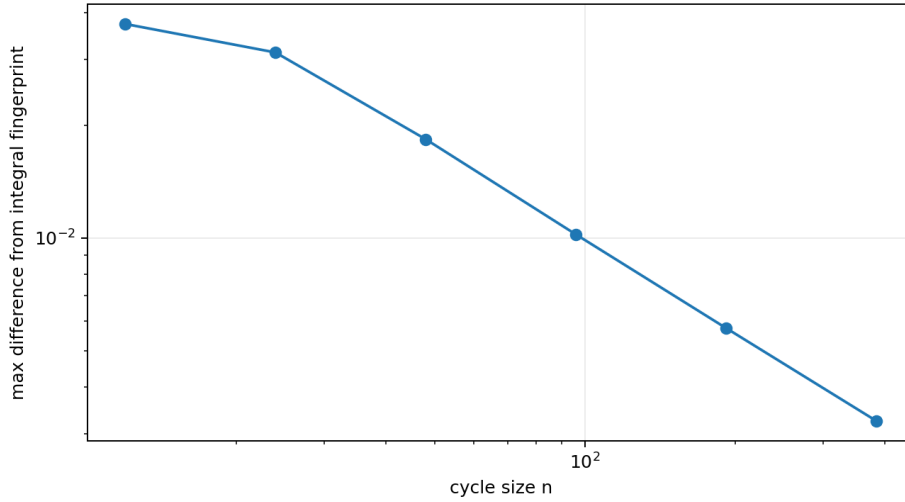


Figure 4: Approach to the declared polyphase integral fingerprint. The omitted infinite weight ℓ^1 tail in the C_{131072} proxy is bounded by 3.51×10^{-4} ; this is not itself a bound on the nonlinear normalized fingerprint error.

The discrepancy decreases monotonically from C_{96} onward. This is strong convergence evidence for this particular envelope and context, not a universal fractal theorem.

8 P502 — exact radial passivity and signed PSD

Along the linear parameter homotopy from Legacy* to strict, the amplitude and denominator remain positive. Hence the sign of the d th radial weight is the sign of

$$\cos((1-t)(\pi d/4 + \pi/6) + t(0.18575d + 0.16250)).$$

The last crossing into the final nonnegative sector is at $d = 6$:

$$t_{\text{radial}} = \frac{5\pi/3 - \pi/2}{5\pi/3 - (6 \cdot 0.18575 + 0.16250)} = 0.9257900390444757 \dots \quad (5)$$

It is isolated in

$$[0.9257900390439756, 0.9257900390449756],$$

where the limiting radial weight changes from -1.02×10^{-13} to $+1.02 \times 10^{-13}$.

The signed Laplacian can be PSD before every weight is nonnegative. Auditing all eleven nonconstant Fourier eigenvalues gives the last PSD crossing in mode one:

$$t_{\text{PSD}} \in [0.7008185569041573, 0.7008185569051574].$$

The minimum nonzero eigenvalue changes sign across the bracket. Thus

$$t_{\text{PSD}} < t_{\text{radial}}.$$

This distinction matters: PSD is a collective spectral property; pointwise radial passivity is a stronger sufficient condition. Neither boundary makes the arbitrary linear parameter homotopy a derivation of strict from legacy.

9 P503 — a nonvanishing phase refinement sector

P494 showed that unconstrained scalar states in $\mathbb{C}^{12}$ have no protected winding. P503 adds exactly the missing typed structure.

Definition 9.1. Let $z_x \in S^1$ and

$$\Delta_x = \text{Arg}(z_{x+1}\bar{z}_x) \in (-\pi, \pi).$$

If every edge stays off the branch cut, define

$$\nu(z) = \frac{1}{2\pi} \sum_x \Delta_x \in \mathbb{Z}.$$

Theorem 9.2 (Bounded-update and refinement invariance). *Suppose $|\Delta_x| \leq \pi - \varepsilon$ and an update has a real lift δ_x satisfying $|\delta_{x+1} - \delta_x| < \varepsilon$ on every cyclic edge. Then $z'_x = e^{i\delta_x} z_x$ satisfies $\nu(z') = \nu(z)$. Inserting on every edge the principal-geodesic midpoint also preserves ν and halves every edge increment.*

Proof. The bound prevents a principal-argument branch crossing. Therefore

$$\Delta'_x = \Delta_x + \delta_{x+1} - \delta_x.$$

Summation telescopes on the cycle. Midpoint refinement replaces each Δ_x by two increments $\Delta_x/2$. $\square$

The implementation replayed 1500 bounded random updates on C_{12}, C_{24}, C_{48} with zero failures. It also found a condition-violating update changing winding from 1 to -1 . P503 therefore constructs a mathematically protected phase sector, but the phase field, branch margin and update law are extra axioms. No particle interpretation follows automatically.

10 P504 — minimal synthetic coherence tester

All 36 single contexts of the P495 catalogue were exhausted. Nine individually separate its four declared models. The strongest one-context design is

$$t = 0.55, \quad \rho_0 = |\hat{1}\rangle\langle\hat{1}|, \quad \text{Fourier-basis measurement.}$$

Its minimum pairwise total-variation separation is 0.235223. With 120 shots, the sum of pairwise Bhattacharyya error bounds is 0.012807. A deterministic-seed simulation of 1000 trials per model gave

$$\begin{pmatrix} 1000 & 0 & 0 & 0 \\ 0 & 1000 & 0 & 0 \\ 0 & 0 & 998 & 2 \\ 0 & 0 & 1 & 999 \end{pmatrix}, \quad \text{accuracy} = 0.99925.$$

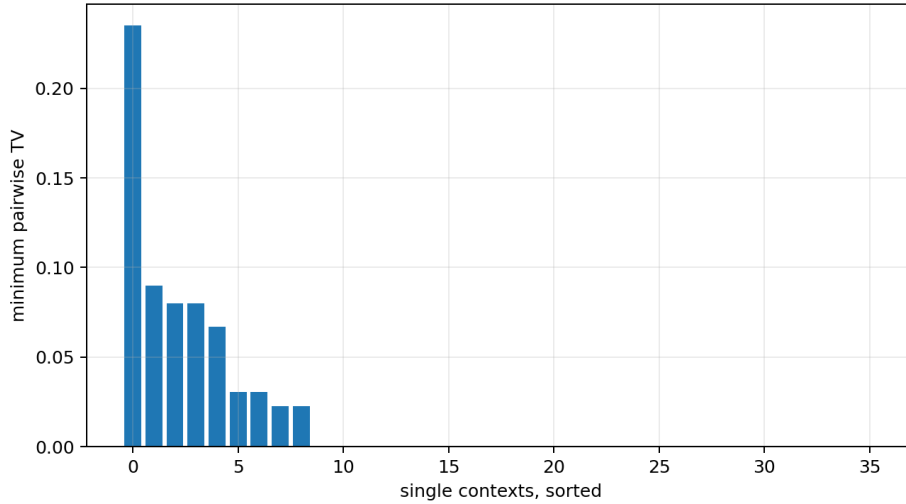


Figure 5: Minimum pairwise total variation for all single contexts, sorted. Only nine contexts separate all four synthetic model families.

Minimality is exact only within this finite catalogue: zero contexts contain no data and one separating context exists. The models, preparation, time, measurement and shot law remain synthetic inputs.

11 P505 — theorem dependency interface

The file `FIN_P497_P506_Theorem_Interface.json` records six theorem nodes:

T488-CYCLE, T489-SCALE, T492-COMPLEMENT,
T493-INDUCTION, T497-IFT, T503-WINDING.

Every node contains assumptions, claim level, conclusion, dependencies, proof skeleton and explicit nonclaims. Schema validation found no missing field; dependency validation found no missing node and no cycle. The canonical payload has SHA-256

79259faf581ecd506124ad1a52465a5fe1e1bc7b95bb8b616ea259598441da81.

The value above is checked against the generated interface during regression testing. This is a proof-assistant-neutral transfer format, not a Lean/Coq/Isabelle machine proof.

12 P506 — parameter-path identifiability no-go

Write

$$K(d; p) = a \frac{\cos(\omega d + \phi)}{1 + \beta d^\eta}, \quad p = (\log a, \log \beta, \eta, \omega, \phi).$$

The Jacobian of $(K(0), \dots, K(6))$ has rank five at the Legacy*, midpoint and strict parameter points. Their respective condition numbers are approximately 1225.8, 49.87 and 544.84. Among five-distance strict designs, the best audited subset is

$$d \in \{0, 1, 2, 3, 6\}, \quad \kappa(J) = 632.75.$$

Thus a static parameter point is locally identifiable after its phase/positivity branch has been fixed.

Theorem 12.1 (Finite-record trajectory no-go). *No observation of a parameter or kernel trajectory at finitely many times $t_1, \dots, t_M$ identifies a unique continuous parameter trajectory in any open admissible parameter set.*

Proof. Let $p(t)$ be one admissible trajectory and choose a nonzero parameter direction v . For sufficiently small ϵ ,

$$\tilde{p}(t) = p(t) + \epsilon v \prod_{j=1}^M (t - t_j)^2$$

is distinct from p but agrees with it at every observation time. Consequently all sampled kernel values agree as well. $\square$

Even a continuously observed unlabelled curve is invariant under monotone time reparameterization. A calibrated continuous time record and a full-rank distance design can recover a path pointwise, but a causal evolution law is still an additional object. P506 therefore rules out deriving the P493 parameter transition from endpoints or any finite set of snapshots.

13 Integrated interpretation

13.1 What has improved

1. **Localization is now mathematically possible.** Strict coupling is compatible with a strongly localized stationary DNLS branch.
2. **The P488 stability signs are certified.** The earlier floating signs reduce to outward-enclosed 2×2 blocks.
3. **The clock question is sharper.** A two-mode clock removes single-period aliasing conditionally on irrationality, and a large finite rational class is excluded.
4. **The Stieltjes fragility is quantified.** The numerical condition number now has a local inverse theorem and an explicit noise threshold.
5. **Approximate context convergence has a target.** The polyphase integral replaces vague fixed-point language by an explicit asymptotic response.
6. **Passivity notions are separated.** Signed PSD and nonnegative radial coupling are demonstrably different boundaries.
7. **The topology obligation is fulfilled conditionally.** An S^1 phase field plus admissible updates supports protected winding.
8. **Operational complexity fell sharply.** One synthetic coherence context suffices within the declared catalogue.
9. **The bridge nonidentifiability is now a theorem.** Static observability does not create a unique parameter evolution law.

13.2 What has not improved

- FIN does not derive the focusing cubic sign, coefficient, or stationary frequency used in P497.
- Numerical continuation to $\kappa = 1$ is not an interval-certified global branch or orbital-stability theorem.
- The localized state is pinned; mobility and collision persistence are untested.
- No absolute time, length, action, mass or energy unit is generated.
- The S^1 phase sector is an added object, not a strict selector or *QW*-2191 discharge.
- The Legacy*-strict homotopy is a diagnostic path, not the missing completion/source theorem.
- Synthetic identifiability is not preparation, apparatus, calibration, raw data or independent custody.

- No legacy physical role is transferred to strict.

Updated scientific interpretation. The strict kernel has now passed a nontrivial capability test: after a standard focusing Hamiltonian nonlinearity is supplied, it supports a strongly localized finite breather. The decisive missing theorem is therefore no longer “can this generator localize at all?” but “does FIN internally select the nonlinear law, its sign and its scale, and can the resulting localized branch be certified and connected to an observable instrument?”

14 Recommended next bounded studies P507–P516

Table 5: Next local analytical and small-compute programmes.

Program	Objective	Acceptance criterion	Priority
P507	Source audit for the focusing nonlinearity	Derive the cubic sign/coefficient from an admitted variational or information functional, or prove that the strict quadratic core leaves it free.	10/10
P508	Certified global DNLS continuation	Cover $0 \leq \kappa \leq 1$ by interval root boxes and overlapping IFT charts; reject any unresolved fold.	10/10
P509	Orbital stability of O201	Apply a finite-graph Grillakis–Shatah–Strauss/Vakhitov–Kolokolov audit and isolate the near-zero Hamiltonian blocks.	9/10
P510	Mobility and Peierls–Nabarro barrier	Continue site- and bond-centred branches, compare constrained energies and test whether a moving localized state can exist.	9/10
P511	Localized spectral-memory dynamics	Couple O201 to the positive Stieltjes state variables of O194 and determine whether localization survives causal memory.	8/10
P512	Strict frequency-ratio arithmetic	Prove irrationality of λ_2/λ_1 or give a precise transcendence obstruction showing why current theorems cannot decide it.	8/10
P513	Certified context convergence rate	Bound the normalized fingerprint error using Wiener-algebra and resolvent estimates rather than an FFT proxy alone.	8/10
P514	Exact signed-PSD homotopy boundary	Interval-certify every Fourier-mode root and prove global PSD/non-PSD on both sides of t_{PSD} .	7/10
P515	Robust one-context tester	Add preparation, timing and POVM uncertainty; retain separation with a worst-case lower TV bound or reject one-context robustness.	9/10
P516	Minimal trajectory-law observations	Under explicit ODE/gradient-flow model classes, determine the smallest time/distance design that identifies a law rather than only a path.	9/10

P507 and P508 are the immediate priorities. P507 decides whether the localization mechanism belongs to FIN or merely to an added DNLS model. P508 decides whether the attractive numerical branch is a theorem across the complete strict coupling interval.

15 Reproducibility

Primary artifacts:

- `fin_programs_497_506_next_research.py`;
- `test_fin_programs_497_506.py`;
- `FIN_Programs_497_506_Results.json`;
- `FIN_Programs_497_506_Summary.csv`;
- `FIN_P497_P506_Theorem_Interface.json`;
- `FIN_Programs_497_506_Figures/`.

Replay commands:

```
MPLCONFIGDIR=/tmp/fin-mpl-1050 \
  python3 fin_programs_497_506_next_research.py
MPLCONFIGDIR=/tmp/fin-mpl-1050 \
  python3 -m unittest -v test_fin_programs_497_506.py
pdflatex -interaction=nonstopmode -halt-on-error \
  FIN_Programs_497_506_Localization_and_Identifiability_Report_EN.tex
```

The main computation uses matrices no larger than 144×144 , a 12-variable nonlinear stationary solve, FFT arrays of length 131072, short interval evaluations and a finite 36-context synthetic search. It is suitable for an ordinary local computer.

16 Final verdict and nonclaims

The P497–P506 batch makes real progress. The strict generator is compatible with localization once a focusing cubic Hamiltonian term is supplied. The localized branch has a theorem-level origin at the anti-continuum limit and a robust numerical continuation to full strict coupling. Several prior numerical observations have been upgraded: P488 stability signs are interval-enclosed, the Stieltjes inverse has a local bound, the passivity threshold is isolated, and the missing topological state space is explicitly constructed.

The falsification boundary is equally important. Static kernel values can identify parameters locally but cannot identify their evolution law. The focusing term is not derived. The phase field is added. The clock remains dimensionless. The operational tester remains synthetic. These facts prevent the localized state from being promoted to experimentally established matter, a universal informational soliton, or a completed physical theory.

No result in this release discharges *QW-2191*, selects an orientation, generates SI or Planck units, completes the Legacy*–strict bridge, transfers a legacy physical role, supplies laboratory evidence, derives the Standard Model or gravity, constructs L_{total} , or closes a Theory of Everything.